

Parting is a sweet sorrow

An Ambient Co-Presence Space
Designed for Adolescents in Transition

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想像一位剛畢業的青少年。

地理上的隔閡，導致沒有跟誰吵架，只是慢慢地、被動地，
和那群朋友失去了聯絡。

當會自然見面的場合消失，友誼就失去現實的聯繫，走向消散。

the study provides strong evidence that most adolescents experience a relatively **brief period of diminished access** to companionship and intimacy with close friends **following relocation**.

The extent of diminution may be greater for adolescents with social anxiety or behavioral concerns.

Vernberg, E. M., Greenhoot, A. F., & Biggs, B. K. (2006)

So What?

Now

孤獨與憂鬱

孤獨,源於**期望的關係**與**實際的關係**之間的落差。

Loneliness emerges from the mismatch between an individual's attained and desired social relations and interactions.

Grigorian, K., Östberg, V., Raninen, J., & Brolin Låftman, S. (2024)

Mid-term

失去對抗社交焦慮的緩衝

好的友誼本身,就是抵禦社交焦慮的保護因子。

Close mutual friendships may help protect socially anxious early adolescents against concurrent psychosocial risks.

Erath, S. A., Flanagan, K. S., & Bierman, K. L. (2010)

Long-term

延續到成年的心理健康

青少年時期的友誼能力,預測了成年後的情緒健康。

Even using this highly conservative approach, adolescent close friendship quality remained a significant long-term predictor of negative affectivity.

Allen, J. P., Costello, M., Hunt, G., Stern, J. A., & Pettit, C. (2024)

WHY current tool not work?

不是有 IG 之類的社交媒體嗎？

這些工具靠「即時＋展演」運作
而展演與比較，正是這個敏感年齡層的**焦慮來源**。

更何況，被動流失的友誼，本就無人主動經營；
而這些工具，全都要求人們主動使用。

fostering rumination on how one is perceived by others, social comparisons and feelings of insecurity about one's social position

(Stritzke, Nguyen, & Durkin, 2004)

青少年既需要**社交臨場感**，否則友誼消散；
又對**評價**極度敏感，而臨場感本身帶來焦慮。

HCI 已經分頭處理過這個矛盾的**不同**部分

每條研究線都解開了矛盾的一半，卻各自留下一道缺口。

第一條線

Ambient Awareness

第二條線

Digital Third Places

第三條線

Asynchronous

HCI RESEARCH GAP

Ambient Awareness

用不打擾的線索,讓人感知他人存在、創造連結感。

→ 幾乎只服務一對一的親密關係。

Digital Third Places

證明數位第三場所能促成低壓力互動。

→ 主題圍繞外部興趣、成員為陌生人,服務興趣社團而非友誼。

Asynchronous

證明非同步能降低社交焦慮。

→ 停在「工具特性」,未系統性地整合進一個互動空間設計。

Peripheral or ambient awareness cues can make users aware of others' presence without interrupting ongoing activities, thereby enhancing a sense of connectedness.

(Dey & de Guzman, 2006)

Providing themed spaces for repeated interactions, supporting user autonomy and customization, facilitating mutually engaging activities, and enabling casual, low-pressure interactions.

(Kim et al., 2025)

Socially anxious individuals show fewer negative interpersonal outcomes in computer-mediated than in face-to-face initial interactions, as reduced cues and asynchronicity ease impression management
(High & Caplan, 2009)

Proposal

一個你和朋友一起搭起的空間、一個帳篷。

鑽進去,裡面放滿**只有你們懂的東西**：

那隻大家一起買給某人的玩偶、某個人常用的筆……

掀開帳篷走出去,是一整片遼闊的星空,

每一顆星,都是一個朋友**來過**的一小朵火花。

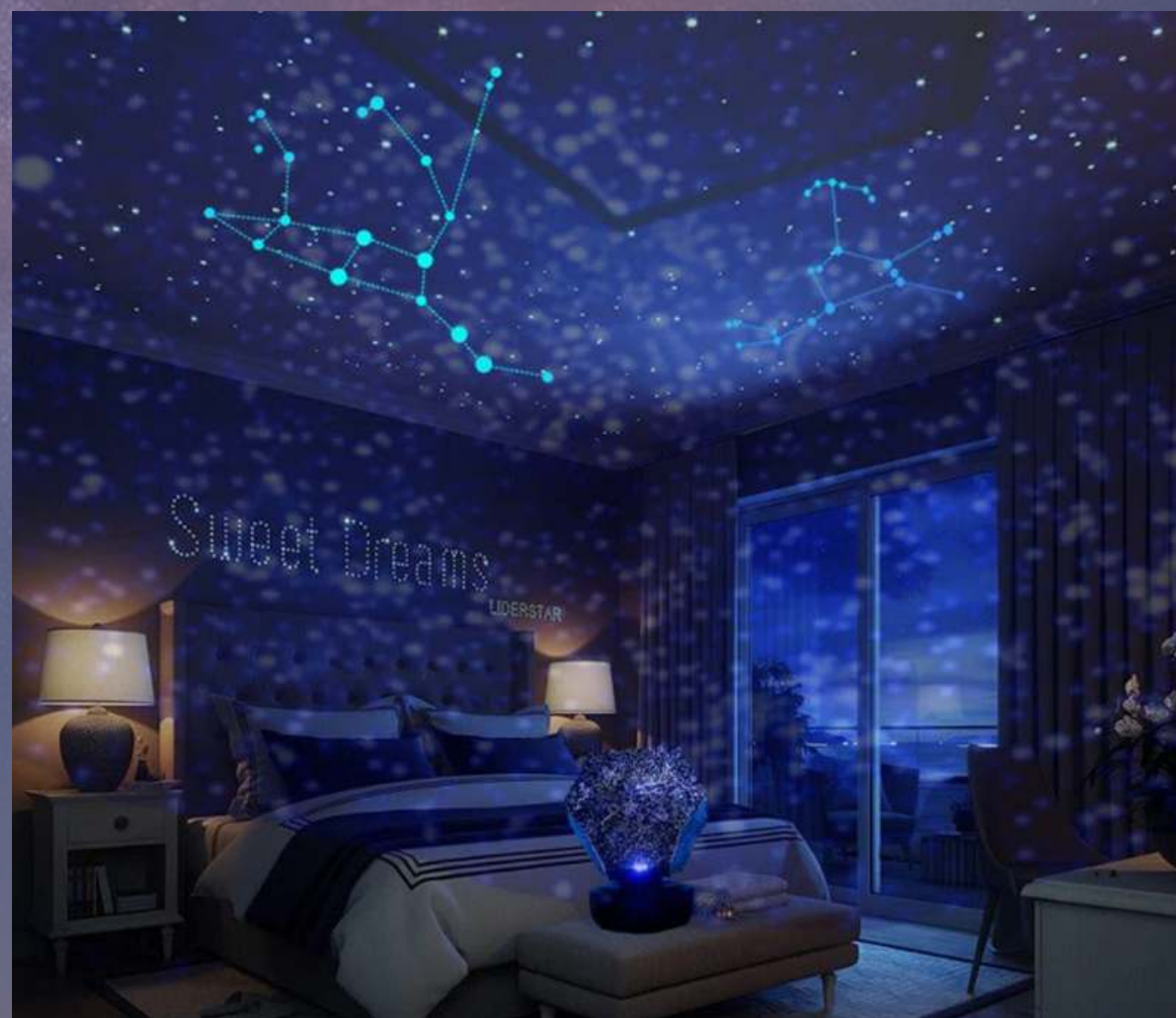
“一個由你們共同記憶的物件構成的、跨距離的環境氛圍裝置。”

Proposal

只有你們懂的東西，以星空為例

一個共享的星空畫布，每個人都可以在上面疊加東西，最終大家看到的是同一片

「被你們共同建造的星空」



3 MECHANISMS

- Agree on interaction vocabulary:
 - light / colorobject typeambient sound
- Design private hints
 - A purple shimmer might mean "thinking of you."
- Add / remove objects
 - Trace design

③ AMBIENT AWARENESS DISPLAY

Digital Family Portrait

Mynatt et al., 2001 · Georgia Tech

Closest to 痕跡

activity level → portrait vibrancy

A photo frame that grows more colorful as an elderly relative is more active that day. No message — just ambient signal of life.

ambient ✓ dyad only no objects

ASTRA

Markopoulos et al., 2004

Used ABC-Q



Family awareness across homes. Proved ambient cues increase connectedness without requiring active communication. Evaluated with early ABC-Q.

ambient ✓ ABC-Q validated

family, not friends

④ TANGIBLE / OBJECT-BASED PRESENCE

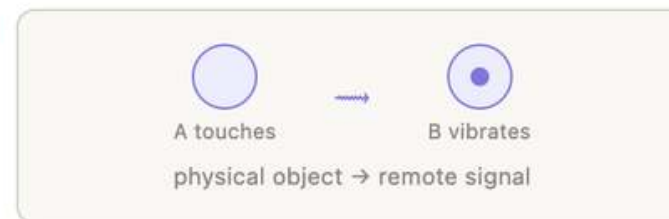
BuddyBeads

Kikin-Gil, 2006

Closest to scan ritual

Jewelry that vibrates when a friend touches their matching piece. No content — just "I thought of you." A low-bandwidth gesture that carries relational meaning.

object-based ✓ no pressure ✓ dyad only



⑤ SHARED PERSISTENT SPACE

Animal Crossing: shared island

Nintendo, 2020 · unintentional third place

Closest analog

Friend groups use shared islands as ambient hangout spaces. Objects placed in the world are traces of presence. Visiting someone's island = entering their tent. Not designed for this, but used this way.

group ✓ object-based ✓ async ✓

game, not designed for this



⑥ INTIMATE CONNECTED APP

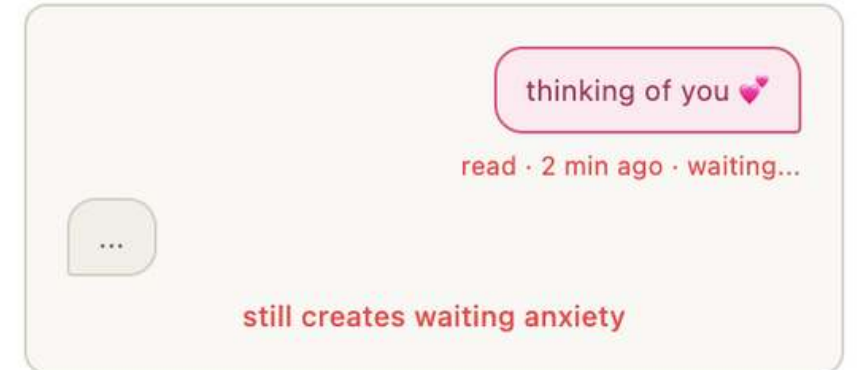
Between / Couple

Commercial apps · 2012–present

Market validation

Private shared space for two people — photo stream, calendar, inside jokes. Shows there's demand for intimate non-public spaces. But still creates response-waiting pressure and only works for dyads.

private space ✓ dyad only × reply pressure ×



⑦ DIGITAL THIRD PLACE

Discord (friend server)

2015–present · informal use

Closest existing group tool

Persistent group space, async-friendly if used softly, has channels for different topics. But fundamentally built on explicit posting — you still have to "perform" by writing something. Public within the group = social comparison possible.

group ✓ persistent ✓ performance-based ×

not ambient ×



GOALS & RESEARCH QUESTIONS

研究目標

提出並探索一個設計,為分散的青少年群體提供臨場感,同時避免展演與比較壓力。

RQ1 (FOUNDATION)

投影式的 Shared space 能否讓人感知到「朋友群體」的存在,而不需直接互動?

RQ2 (CORE)

這種模糊、非展演的臨場感,能否在維持連結感的同時降低展演與比較壓力?

RQ3 (REAL WORLD)

這種空間,如何影響青少年維繫遷移後友誼的意願與感受?

EVALUATION PLAN

RQ1: 帳篷氛圍與星空, 能否讓人持續感知到「朋友群體一直存在」, 而不需直接互動?

測「他們一直在」的感受度, 不測「能感受到多少人」的判斷準確率

RQ2: 這種模糊、非展演的臨場感, 能否在維持連結感的同時降低展演與比較壓力?

採用 ABC-Q 量表, 量測「連結感收益」與「壓力成本」

→ 檢驗能否做到「收益不降、成本下降」

Sharespace vs 現有工具

- 情感效益 Affective Benefits

- 情感成本 Affective Costs

- 知覺與覺察 Perceived Awareness

RQ3: 這種空間, 青少年能否持續維繫遷移後的友誼?

質性訪談: 他們是否覺得, 靠這種不必經營形象的在場, 更能、也更安心地和朋友保持連結? 怎麼發生的?

the development of a new questionnaire measure, the Affective Benefits and Costs in Communication Questionnaire (ABC-Q), that is aimed to assess the affective characteristics of communications

(IJsselsteijn et al., 2009)

Thank You

